

Lab meeting: HA next steps

Recalibration after Roro reply and CI note

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17 July 2026

Patrick Manson Building, 1/F — 9:30–11:30

Purpose

- ▶ Align on the **next HA analysis** after Roro's confirmation and Prof. Bishai's instruction.
- ▶ We do **not** yet have HA results — access still requires wet-ink signatures and HPC onboarding.
- ▶ Ask the room to lock a short list of decisions so merge work can start immediately after access.

Recalibration (CI direction)

We adjust to the CI note:

1. Merge monthly temperature (**Tmax**, **Tmin**, and **lag**) to monthly AMI and stroke admission risk.
2. Covariates: age, sex, diuretics, beta blockers, metformin, SGLT2is, BMI.
3. Next write-up: **Table 1** with *N*, means, and SDs for temperature and covariates.

Official hot nights / cold days remain available as **secondary** exposures — not the near-term primary story.

What Roro locked (12 July)

- ▶ Coding: **ICD-9** (DAE records included — define DAE today).
- ▶ Grain: **patient-level** complete admissions.
- ▶ ED-to-inpatient: **not in this extract**.
- ▶ Age bands: stratification supported, including **65–69** and **70–74**.
- ▶ Access: wet-ink signatures from **Bob** and **Hogan**, then add members to the **HPC** server.
- ▶ Open: whether an **IRB protocol update** is needed (PI decision).

Ready now (public side)

- ▶ Monthly HKO file for 2013–2023 exists ($N = 132$ months): mean, T_{min}, T_{max}, extremes, spells.
- ▶ Annual extremes validated 33/33 against HKO *Year's Weather*.
- ▶ C&SD mid-year age×sex denominators are in the repo.
- ▶ **Blocker:** HA patient file not yet available to Bob — merge starts on HPC after signatures.

Table 1 shell (temperature filled; covariates pending)

Variable	<i>N</i>	Mean	SD
Mean temperature (°C)	132	24.02	4.70
Mean T _{min} (°C)	132	22.10	4.69
Mean T _{max} (°C)	132	26.66	4.82
Hot nights (days/month)	132	3.40	5.40
Cold days (days/month)	132	1.10	2.55
Very hot days (days/month)	132	3.19	5.18
Age, sex	—	<i>pending HA</i>	—
Diuretics / beta blockers	—	<i>pending HA</i>	—
Metformin / SGLT2is	—	<i>pending HA</i>	—
BMI	—	<i>pending HA</i>	—

Source: data_processed/climate_monthly_2013_2023.csv (HKO Headquarters). HA covariate rows after server access.

Decisions for the room

1. **IRB:** update protocol for this HA use? (PI)
2. **DAE:** what does “DAE records included” mean?
3. **Principal diagnosis:** available / required?
4. **Clinical fields:** are diuretics, beta blockers, metformin, SGLT2is, BMI on the approved extract?
5. **Lag:** is lag-1 month the default for temperature?

This week's actions

Owner	Action
Bob + Hogan	Wet-ink signatures for HA/HPC access
Roro	Onboard Bob to HPC after signatures
Bishai	IRB call; confirm covariate set / lag default
Bob (after access)	Merge T _{max} /T _{min} (+ lag) to AMI/stroke; draft Table 1
Yuchen	Calendar invite (lab meeting logistics)

Next scientific deliverable: Table 1 + first temperature–admission associations on HPC.